

High Resolution Position Estimation from Hall Effect Sensor

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Abstract—This paper provides an alternative method for high resolution position estimation of a BLDC motor by obtaining pulses from the hall effect sensor. This method proposes to use a fast processing time in order to achieve the best performance. In the document the development of the methodology is shown for obtaining the high resolution position. The problem is the low resolution and it is due to the hall effect sensor that only provides maximum 12 rising and falling edges in the three sensors, thus the position comes to be 12 pulses per revolution. The method is demonstrated through a simulation using the considerations for fast processing.

I. INTRODUCTION

In the industry, the motion control systems are driven by DC or AC motors. Moreover, the sensors attached to the motors can varies between optical incremental encoders, effect hall sensor and resolvers for motor position. However, the sensors does not provide the motor position directly; in many applications, it is necessary apply digital processing to the sensors signals.

For instance, for optical incremental encoder there are many methods in the literature [1]–[4]. The M, T, S, M/T and N methods had been applied in order to obtain the motor position. Every method has advantages and disadvantages, but there are many factors that affects the performance of the methods. For consequence, the sampling time is a factor that can changes from the motor position resolution. Nowadays, the conventional processing platforms can process at different processing time. However, the processing platforms with sequential programming usually takes several cycles clock to execute an instruction and the quality signal is affected.

On the other hand, brushless DC motor (BLDC) includes hall effect sensors for motor position. This kind of sensor produces only 12 edges per revolution, because the sensors are in rotational arrange with 120 degree between them, as shown in Figure 1. The hall effect sensor has many advantages versus the optical incremental encoder. Another advantage offered by the hall effect sensors on other positioning sensors is their resistance to various factors such as grease, dust, air and RF [5] noise. In addition, there are no sparks and it is more reliable. However, 12 pulses per revolution is a very low resolution to determine the motor position.

In the literature, the BLDC motor has been applied in many fields and there are several methods for controlling the position of BLDC motors. For example, using high performance sensors such as incremental encoders with sinusoidal output

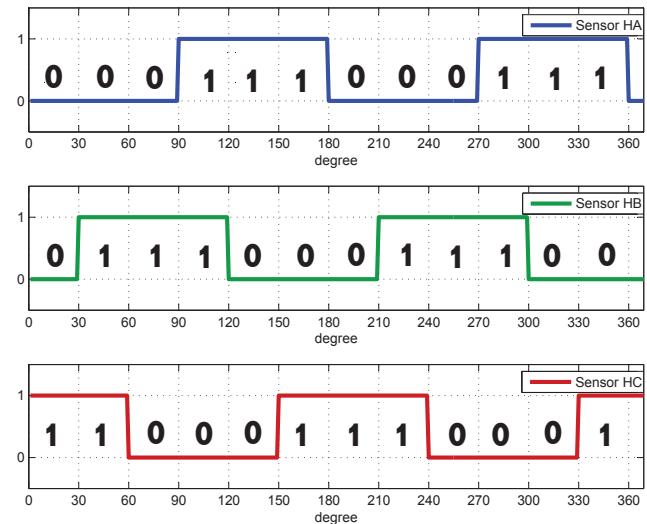


Fig. 1. Hall effect sensor signals of 360°.

position. But this type of sensors not only increase the cost and size of the system, also reduce their reliability [6], [7].

As an alternative to using high performance sensors on the market can be found in motors including its structure hall effect sensors to aid in positioning control. The use of hall effect sensors require a minimum cost and volume compared to other high-resolution sensors [8]. The methods of control for BLDC motors require the use of high-resolution sensors that besides increasing the cost and size of the system, reduce their reliability [7]. To estimate position hall effect sensors bring a reduction in the cost and volume in the system when compared with their counterparts in high resolution [9]. The motors on the market have mostly an internal arrangement of hall effect sensors that allow the switching of the phases with which the motor operates. The sensors provide signals that give the position of the motor shaft with a resolution of 30 degrees [8]. It is in this low resolution that lies the need to adapt sensors with higher performance.

In commercial devices you can be found usually three hall effect sensors positioned in a specific manner within the motor structure. The signals from the hall effect sensors are analog signals and can provide discrete information of the absolute position of the motor with a resolution of 60 electrical degrees [6]. The error in the resolution should be reduced in order to

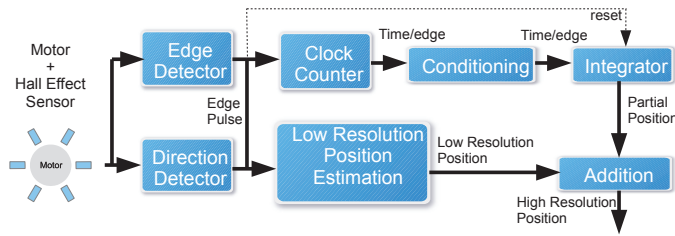


Fig. 3. Block diagram of the method to estimate high resolution position.

make movements with the precision required to reduce the error is necessary to apply signal processing by converting these analog signals to digital.

The processing of analog signals to estimate the position of the motor provides information that allows implementing control algorithms and programmable estimation algorithms by using devices such as integrated circuits, microcontrollers and field-programmable gate array (FPGA).

Morimoto et al. [6] they proposed a traction system for a motor sine wave permanent magnet by means of position sensors low resolution. Morimoto and his colleagues obtained information from high-resolution low resolution sensors through a position estimator circuit. The results were compared with a position sensor sine wave and the authors concluded that the results of the proposed method was similar to those obtained by high resolution sensors except the time of acceleration and deceleration of the motor.

Lidozzi et al. [7], implemented a system based on the electromotive force in conjunction with three hall effect sensors to estimate a rotor position of permanent magnet synchronous motors for traction in a wheelchair method. The authors used a speed estimation method based on the model of the motor frame and the position calculated by integrating the estimated speed.

Liaw et al. [9], designed and implemented a robust position control for linear brushless motor. Position control was divided into different parts. First, they used hall effect sensors and linear encoders for position information of mobile member.

Bu et al. [10], presented a control scheme for synchronous machines operated sinusoidal manner. In his work they implemented the equation of mechanical motion to an observer to estimate rotor position every 60 electrical degrees restarted. The observer was assisted by hall effect sensors to ensure that the performance of the control algorithm was similar to that obtained for a system with a position sensor higher resolution.

Morales et al. [11] simulated and implemented a control algorithm based on field-oriented Clarke transform into a permanent magnet motor. The rotor position was estimated by three hall effect sensors. The authors obtained a greater resolution of the position by the dynamics of the rotor.

The research presented by Simpkins et al. [12] a new method was presented to detect the absolute position of a BLDC motor using an analogy in the hall effect sensors instead of a binary value signal. The use of sensors with sinusoidal output signals allows the estimator developed have a count of 4932 states per revolution, giving a resolution of 0.073 degrees.

TABLE I. HALL EFFECT SENSOR PHASES OF 180°

Phase	Sensor A	Sensor B	Sensor C
1	0	0	1
3	0	1	1
2	0	1	0
6	1	1	0
4	1	0	0
5	1	0	1

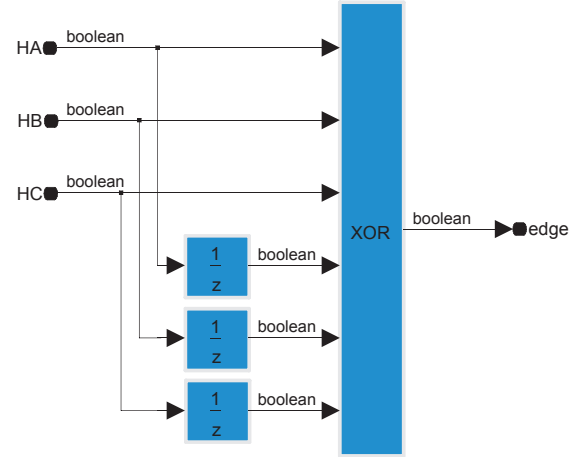


Fig. 5. Block diagram to detect the rising or falling edges.

To achieve optimum reliability in this system is necessary to know the precise location of the sensors inside the motor, otherwise the appropriate electronic switching motor torque submit unwanted variations.

Alaeinovin and Jatskevich [13], proposed algorithm to filter the signals obtained by the hall effect sensors to improve the operation of BLDC motors. To filter the signals obtained by the hall effect sensors authors evaluated the frequency of the signals obtained by Fourier series of discrete time.

Ozturk et al. [14] presented a direct torque control for permanent magnet synchronous motor using hall effect sensors. The authors estimated the position and speed using three hall effect sensors which provided location information every 60 electrical degrees. The estimation of the rotor position and speed values in each interval of 60 electrical degrees was obtained by the equations of Newton's law of motion and error correction was used giving a logical values to the position according to certain conditions in grades.

Haace and Franc [15] proposed a method for estimating position of a BLDC motor for bilateral teleoperation system using hall effect sensors located inside the motor frame. The authors took the three signals from the hall effect sensors and transformed into two phase signals by Clarke transform.

II. POSITION ESTIMATION FROM HALL EFFECT SENSOR

In order to control a BLDC motor, it is necessary to sense the position signal. The hall effect sensor provides a analog or discrete signal that can be used to determines the position. However, the low resolution affects the performance. In this paper it is proposed a alternative method to obtain a high resolution position by using a hall effect sensor as shown in

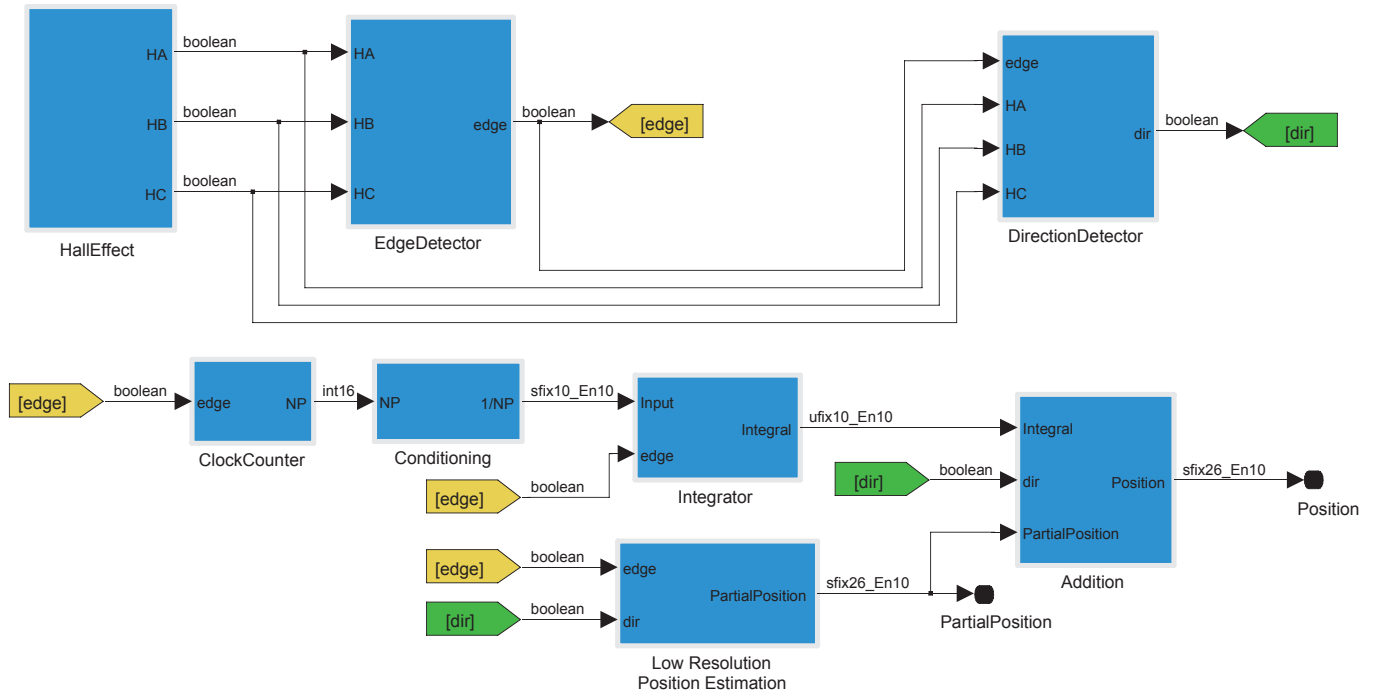


Fig. 2. Simulink block diagram of high resolution position estimation from hall effect sensor based on FPGA.

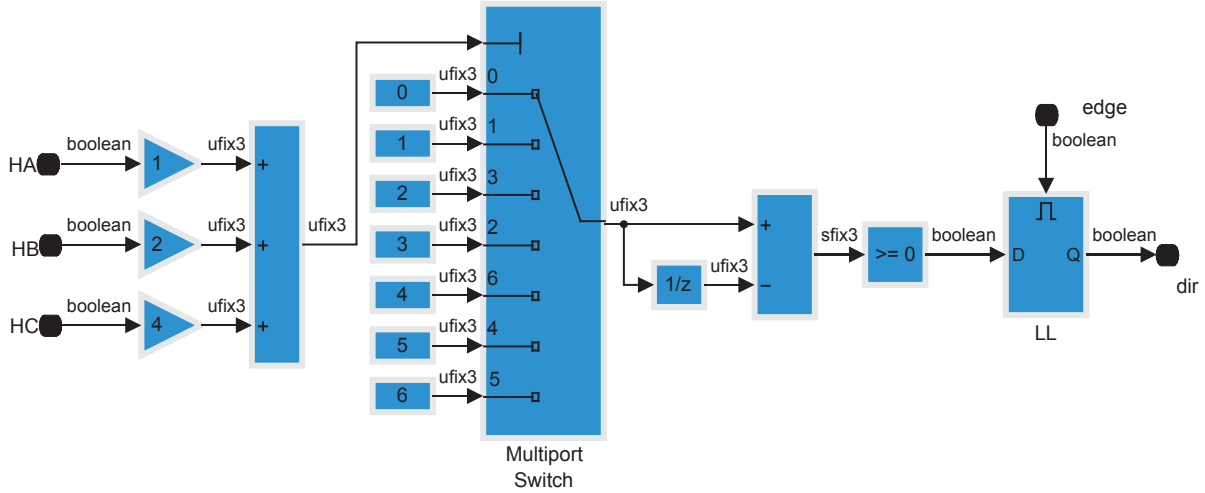


Fig. 4. Block diagram to determine the direction.

Figure 3. The proposed method include edge detector, direction detector, clock counter, conditioning, integrator, low resolution position estimation and addition blocks.

Hall effect sensors have three discrete signals of the sensors 120 degrees apart circular array. These sensors generate signals as a whole show the steps of detecting magnetic field. The phases that indicate the rotational position can produce a maximum of 6 phases. The binary codes generated in each motor phase detects the state in which the is present. The sensors never produce 000 or 111 as shown in Figure 1. The phases shown in the Table I lets to detect the rotation direction and the change of phase called edge.

Figure 2 shows the Simulink block diagram with the

interconnection between blocks. The *Position* is the high resolution position signal. All mathematical operations are established in fixed point format. The format helps convert fixed point algorithm into other languages such as VHDL or Verilog for FPGA. The widths of words in fixed point operations were selected depending on the range and the precision desired in the output of operations. Table II shows the data type for calculations in the mathematical operations. The output *Position* is *sfix26_En10*, which means that is possible to obtain position of 2^{16} pulses with partial steps of $1/2^{10}$ pulses. The proposed method is based on calculating the duration between each edge and get the inverse of the time, that is, obtain the frequency of the pulses. The pulse frequency is in other words, the rate at which pulses are produced per

TABLE II. DATA TYPE OF CALCULATIONS IN THE ALGORITHM ESTIMATION.

Data Type	Signed/Unsigned	Width	Integer	Precision
<i>boolean</i>	Unsigned	1	1	1
<i>uint16</i>	Unsigned	16	16	1
<i>int16</i>	Signed	16	16	1
<i>sfix3</i>	Signed	3	3	1
<i>sfix16_En16</i>	Signed	16	0	$1/2^{16}$
<i>ufix10_En10</i>	Unsigned	10	0	$1/2^{10}$
<i>sfix10_En10</i>	Signed	10	0	$1/2^{10}$
<i>sfix26_En10</i>	Signed	26	16	$1/2^{10}$

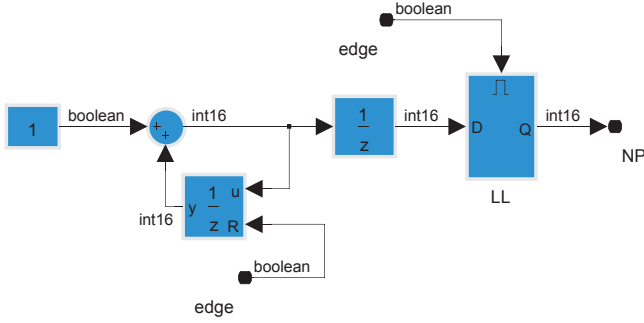


Fig. 6. Block diagram of pulse clock counter.

second. Obtaining the speed, it can be integrated to obtain the position again, in this case with a differential time equivalent to the processing time.

Direction detector block: The dir detector block determines the rotation direction, the point is to obtain the subtraction of the present and the last sensor phase. The rotation direction is updated every edge pulse as shown in Figure 4. Initially the sensor signal is converted to a decimal number using *HA* as least significant bit and *HC* as most significant bit with a *sfix3* data type. The number acts on the multiport switch and generates an ordered sequence of phases also with *sfix3* data type. Then, the difference between the current phase and the previous phase is obtained with *sfix3* data type. However, the difference when the phase 1 minus 6 is equal to -5, but the output is reduced only generates 3 due to the *sfix3* data type. Then, a sequence of positive or negative numbers according to the direction of rotation is obtained. These pulses are generated by only one clock cycle, which is maintained through latch.

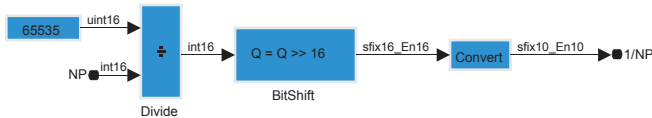


Fig. 7. Block diagram of number of pulses conditioning.

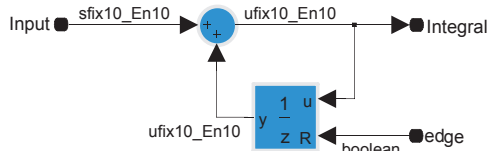


Fig. 8. Block diagram of integrator.

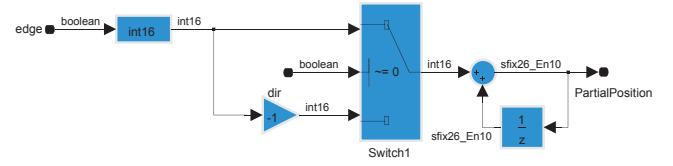


Fig. 9. Block diagram of low resolution position estimation.

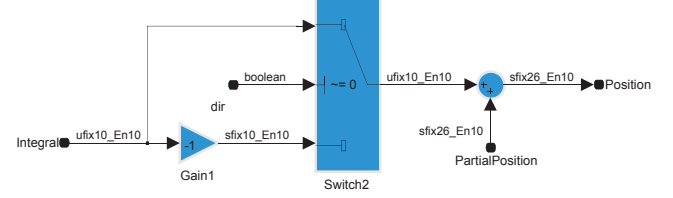


Fig. 10. Block diagram of addition of low resolution plus partial position.

The data type for direction is *boolean*.

Edge detector block: The edge detector block defines the output *edge* that produces a single pulse every change of phase with a XOR bit operator of present and past sensor state as shown in Figure 5. All the mathematical calculations are boolean data type because the hall effect sensor provides only two states 1/0.

Clock counter block: The clock counter block is a clock pulse integrator that is restarted every edge pulse. In order to obtain the number of clock pulses to obtain the duration between edge pulses, the count is captured every edge pulse also as shown in Figure 6. Practically, the number of clock pulses can determines the time between edge pulses.

Conditioning block: Essentially, this block calculates the inverse number of clock pulses obtained in the previous stage. The goal is to obtain differential amount that can be integrated each clock pulse. The amount is like a fraction of pulse of low resolution position. Figure 7 shows the mathematical operations. Firstly, 65535 is divided by the number of pulses NP. Then, a bit shift operator is carried out to move the decimal point 16 position to the left. Mathematically, the calculation is represented in the equation 1.

$$\frac{1}{NP} = \frac{2^{16}}{NP} * 2^{-16} \quad (1)$$

where 2^{-16} is performed as a bit shift operator to the left.

Integrator block: In order to obtain the accumulated partial position, this block integrates every clock pulse the amount obtained in the previous stage as shown in Figure 8. The maximum integral value is the unit, that represent only one pulse. If the unit is reached, the integration stop and remain in 1 until the next edge pulse. The output resolution is 10 bits that means $1/2^{10}$ of precision.

Low resolution position estimation block: The low resolution position is a counting of edge pulses. Each edge pulse is generated every 30 degrees, that is only 12 pulses per revolution. The quantization error is up to 30 degrees. For motion control application this resolution is not attractive and

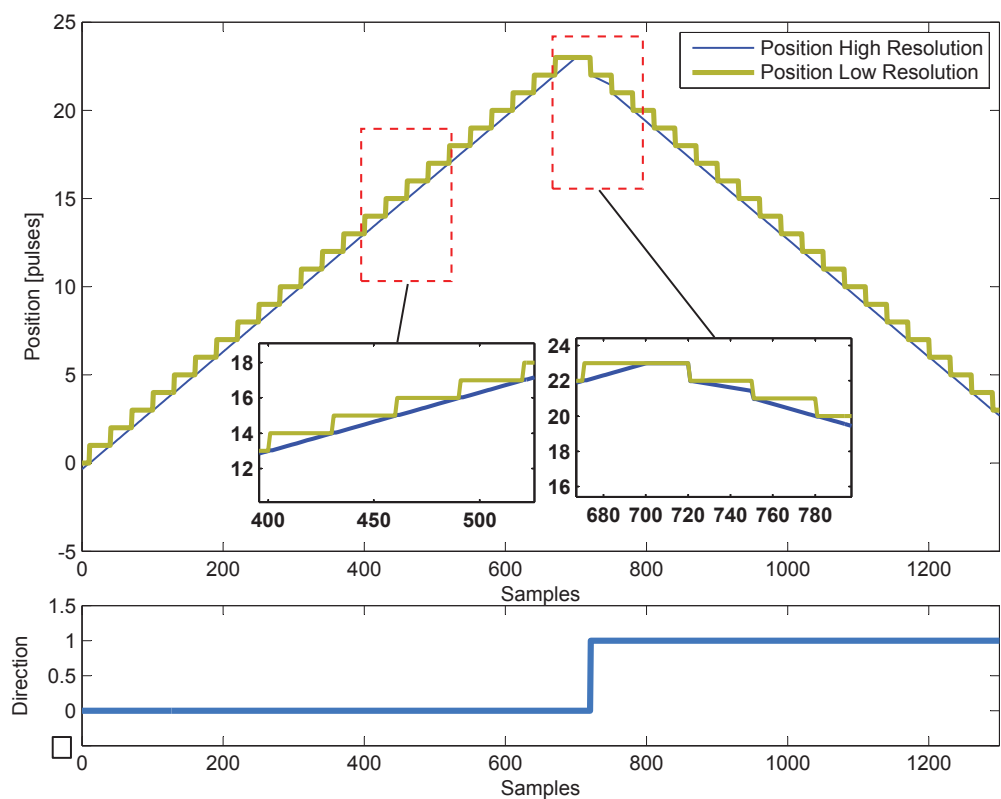


Fig. 11. Simulation result of high resolution position.

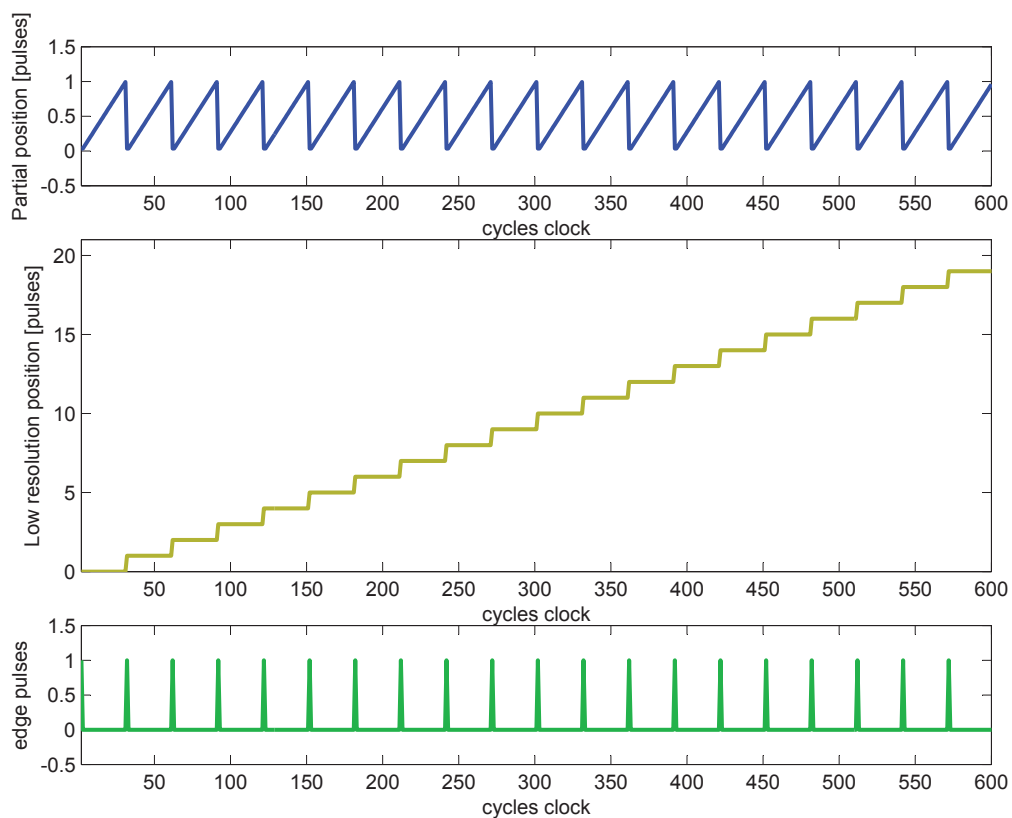


Fig. 12. Simulation result of partial position, low resolution position and the edge pulses.

the performance is deteriorated. Figure 9 shows the integration of edge pulses.

Addition block: This block is responsible for carrying out the sum of the low resolution position more partial position corresponding accumulated in each clock cycle. In this case the data type of the position of low resolution is 16 bits and resolution of accumulated partial position is 10 bits. Therefore the high resolution position is 16 integer and 10 decimals which is represented by *sfix26_En10* as data type. The calculation is simple addition with direction (sign) detection, as shown in Figure 10.

III. SIMULATION RESULTS

The high resolution is achieved with the proposed method, as shown in Figure 11. It is possible to verify in the detail that the points are joining the discrete low resolution position. The proposed method reach the maximum error when the direction is changed. However, the error is not more than the low resolution position.

Meanwhile, Figure 12 shows the partial position that is equivalent to the quantization error of low resolution position. In addition, it is possible to verify the edge pulse occurrence in the position estimation.

IV. CONCLUSION

In this paper has been described an alternative method for high resolution position from hall effect sensors. The low resolution that can be obtained from the only 12 pulses per revolution has been enhanced for motion control applications with BLDC motors. The resolution depends number of bits and on the cycle clock of the processor used for experimental applications.

The mathematical operations are performed with fixed point calculations, that is adequate for FPGA programming. Moreover, the fixed point calculation can be executed at faster processing than floating point calculations.

REFERENCES

- [1] T. Tsuji, M. Mizuochi, H. Nishi, and K. Ohnishi, "A velocity measurement method for acceleration control," *IECON, 31st Annual Conference of IEEE Industrial Electronics Society*, 2005
- [2] S. H. Lee and J. B. Song, "Acceleration estimator for low-velocity and low-acceleration regions based on encoder position data," *IEEE/ASME Trans. on Mechatronics*, vol. 6, no. 1, pp. 58-64, 2001
- [3] H. Tanaka, K. Ohnishi, and H. Nishi, "An approach to velocity estimation using FPGA," *IECON, 33rd Annual Conference of the IEEE Industrial Electronics Society*, 2007
- [4] M. Nandayapa, C. Mitsantisuk, and K. Ohishi, "High resolution position estimation for advanced motion control based on FPGA," *IECON 2012 - 38th Annual Conference on IEEE Industrial Electronics Society*, Montreal, Canada, pp.3808,3813, 25-28 Oct. 2012
- [5] J.M. Cesaretti, "Mechanical stress and compensation in hall sensors," M.Sc., *Georgia Institute of Technology, USA.*, 2008.
- [6] S. Morimoto, M. Sanada, and Y. Takeda, "Sinusoidal current drive system of permanent magnet synchronous motor with low resolution position sensors," *IEEE Proceedings IAS Annual Meeting*, San Diego, CA, pp. 9-13., Oct. 1996.
- [7] A. Lidozzi, L. Solero, F. Crescimbeni, and A. Di Napoli, "SVM PMSM drive with low resolution hall-effect sensors," *IEEE Transactions on Power Electronics*, vol. 22, no. 1, pp. 282-290, Jan. 2007.
- [8] K. Sam-Young, C. Chinchul, L. Kyeongjin, and L. Wootaik, "An improved rotor position estimation with vector tracking observer in PMSM drives with low resolution hall-effect sensors," *IEEE Transactions on Industrial Electronics*, vol. 58, no. 9, pp. 4078-4086, Sep. 2011.
- [9] C. M. Liaw, R. Y. Shue, H. C. Chen, and S. C. Chen, "Development of a linear brushless DC motor drive with robust position control," *IEEE Proceedings-Electric Power Applications*, vol. 148, no. 2, pp. 111-118, 2001.
- [10] J. Bu, L. Xu, T. Sebastian, and B. Liu, "Near-zero speed performance enhancement of PM synchronous machines assisted by low-cost hall effect sensors," *Applied Power Electronics Conference and Exposition*, 1998. APEC '98. Thirteenth Annual Conference Proceedings 1998., vol.1, no., pp.64-68, Feb. 1998.
- [11] D. A. Morales, K. Findlater, and V. Chandran, "A motor controller using field oriented control and hall effect rotor position sensors : simulation and implementation," *IEEE Proceedings of the 6th European Embedded Design in Education and Research*, pp. 235-239, Milano, Italy, 2014.
- [12] A. Simpkins y E. Todorov, "Position estimation and control of compact BLDC motors base on analog hall effect sensors," *American Control Conference*, pp. 1948-1955, Baltimore, EEUU, july 2010,
- [13] P. Alaeinovin and J. Jatskevich, "Hall-sensor signals filtering for improved operation of brushless DC motors," *IEEE International Symposium Industrial Electronics Society*, pp.613-618, June. 2011
- [14] S. B. Ozturk, B. Akin, H. A. Toliyat and F. Ashrafzadeh, "Low-cost direct torque control of permanent magnet synchronous motor using hall-effect sensors," *IEEE Proceedings Applied Power Electronics Conference*, pp.19-23, March 2006.
- [15] A. Hace and M. Franc, "Pseudo-sensorless high-performance bilateral teleoperation by sliding-mode control and FPGA," *IEEE/ASME Transactions on Mechatronics*, vol. 19, no. 1, pp. 384-393, 2014.